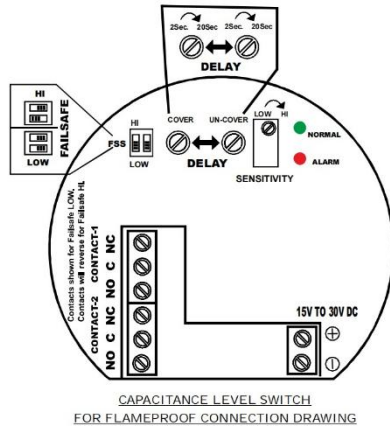


Capacitance Level Switch Operation Manual

In capacitance level switch sensitivity adjustment is carried out on all models before shipment and there is no need for the customer to perform this task by himself or herself unless required.

※ When there are materials or mixtures with multiple characteristics such as high/low dielectric and conductive characteristics simultaneously (for example; silica sand in the glass industry, etc), it must be tested on the site first, to determine whether the sensor is suitable or not. If there still exists measuring problems, we suggest using our Vibrating Fork Level Switch (Solid/Liquid) or Rotating Paddle Level Switch.



INSTALLATION

- (1) If installed multiples, The distance between the two-level switches should be at least 300mm and the distance between the side of the silo and top-mounted switch should be at least 300mm.
- (2) The grounding sleeve should protrude more than 30mm from the side of the silo.
- (3) The side-mounted probe should be mounted at angle of 45°
- (4) Conduit faces downward at installation.
- (5) used outdoors, the conduit opening should point in a downward direction to prevent rainwater from entering the device.

If the capacitance level switch is mounted on the tank, users should ensure that the tank / silo is completely empty before the initial test is carried out.

- (1) Users should ensure that the correct power 24V DC has been given according to model.

Preset for Sensitivity Adjustment is on right side of RED and green LED's.
Can be used for further tuning.

CALIBRATION (Sensitivity Adjustment)

INITIAL CALIBRATION

1. After installation with power supply, make sure no material within 300 mm around the probe.
2. Turn the "SENSITIVITY PRESET" Clockwise position using micro flat head screw driver till the RED LED GETS ON.
3. Using flat-head screwdriver, turn the Preset Anti clockwise for adjustment until GREEN LED Gets ON and RED OFF

Initial calibration is complete now. This step is only required in fine tuning cases.

REPEAT SENSITIVITY ADJUSTMENT

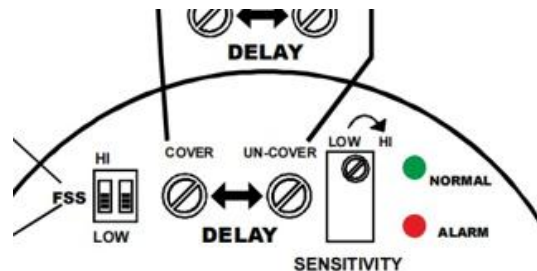
1. Make sure the "RED LED" sign does not light up, when the medium does not contact with the probe, and vice versa when the medium contacts or covers the probe, then the "RED LED" sign lights up.
2. Gradually, adjust the preset clockwise direction until "GREEN LED" light turns off.(RED LED ON)
3. turn the preset counterclockwise back for half of the recorded turning angle till RED LED light gets OFF.(GREEN LED GETS ON)
4. For lower the sensitivity, turn the preset towards counterclockwise.

DELAY ADJUSTMENT

- (1) Cover and uncover, both delays are provided. The user can set the delay according to requirements.
- (2) To increase the delay time, use a screwdriver to turn the "delay" trimmer clockwise. To decrease the delay time, turn the "delay" trimmer counterclockwise.

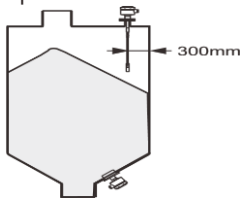
CONTACT OUTPUT

- (1) If the probe is not in contact with the material or the bin is empty, the RED LED will not switch on. NC and COM contact points will remain connected. 2 set of c/o contacts. Refer Drawing.
- (2) If the probe is in contact with the material, the RED LED will switch on without delay. NO and COM contact points will come into contact.



INSTALLATIONS

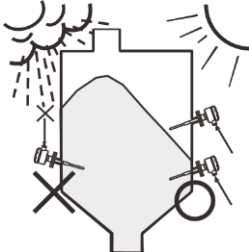
The insulation part must be 30cm inside the tank at least to ensure sensitivity is adjustable and to avoid false action when medium build up.



For top mounting, angle of repose must be considered when installation. Flexible Wire Probe Model must be installed min. 300mm from tank wall. Disc type Plate Model is applicable for bottom and or side mounting.

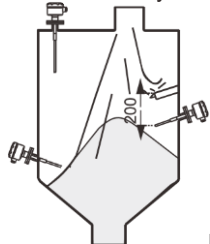
Cable entry must face down firmly locked to avoid rain damage.

Plate Model installation,

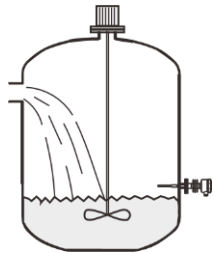


thickness including tank wall & connection flange must be less than 25mm to avoid false action.

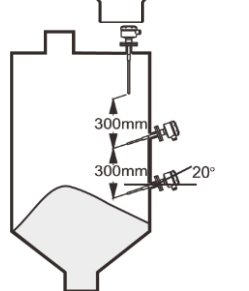
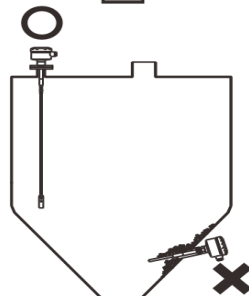
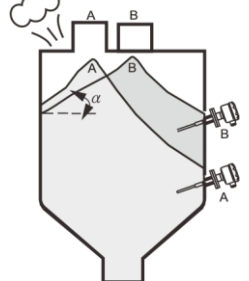
It's recommended to install the switch away from



inlet to avoid damage. If switch must be installed near inlet area, a protective cover must be placed 200mm on top of the switch.



Top mounting is recommended for medium with poor flowability to avoid medium build up. overflowed

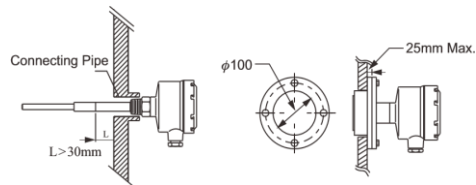


Be cautious with installation location. Angle of repose must be considered if inlet is not located in center top of the tank.

When multiple switches are installed, keep at least 300mm distance for each switch. Mounting the probe at a 20° incline will

optimize the results and increase sensitivity. It also won't be damaged by the inflowing material.

Switches with delay time function is recommended for tank with agitator to avoid false action.



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